

Assignment 2: Panel Data – Temperature Shocks and Economic Growth

Advanced Methods for Data Analysis in Public Policy Research FS 26 — ETH Zürich

2026-03-23

Application Background

This assignment is based on Dell, Jones & Olken (2012), who study how year-to-year variation in temperature within countries affects economic growth. The key idea is to use a panel model with country fixed effects to isolate the impact of temperature shocks from time-invariant country characteristics.

Research Question: How do temperature shocks affect GDP per-capita growth, and do effects differ between poorer and richer countries?

Instructions

- Provide all code and answers within the provided markdown template.
- Ensure that you formulate a written answer where asked. Be concise.
- **Pass requirements:**
 - The analysis must be replicable on any machine. Use relative file paths when loading the data.
 - Completion of all questions is mandatory for a passing grade.
- **When you are done:** Submit the Markdown script **and** the compiled HTML file on Moodle.
- **Submission deadline:** 30.03.2026, 9:00 am.
- **Presentation:** The assignment will be discussed during the exercise session on 13.04.2026. Be prepared to present the assignment in class. Every group member must be able to discuss and defend any part of the assignment.

Exercises

1. Panel structure

- a) Is the dataset a balanced panel? Explain how you check.
- b) Report the number of countries and the average number of years observed per country.

2. Descriptive statistics

- a) Report summary statistics for *gdp_growth* and *temp_c* (min, max, mean, sd; and selected percentiles).
- b) Pick one country and plot *temp_c* over time. Briefly describe the pattern.

3. Pooled regression with clustering

- a) Regress *gdp_growth* on *temp_c* and *precip_mm* using pooled OLS. Use standard errors clustered at the country level. Interpret the coefficient on *temp_c*.
- b) Explain why clustered standard errors at the country level are appropriate for this specification.

4. Country fixed effects

- a) Estimate a fixed effects model with country FE:

$$gdp_growth_{it} = \alpha_i + \beta temp_c_{it} + \gamma precip_mm_{it} + u_{it}$$

- b) Compare β to the pooled estimate. Why might they differ?
- c) In one paragraph, explain what country FE control for.

5. Two-way fixed effects

- a) Add year fixed effects to the model in (4). Compare the temp coefficient to (4).
- b) Explain what year FE absorb in this context.

6. Temperature ‘shocks’ and heterogeneity

Dell et al. emphasize within-country shocks and heterogeneous effects by income.

- a) Regress *gdp_growth* on *temp_dev* and *precip_dev* with country FE and year FE.
- b) Add the interaction term *temp_dev_poor*. Interpret the coefficient on *temp_dev_poor*.

7. Robustness / influential units

- a) Identify one or two potentially influential countries and re-estimate your preferred model excluding them. Do results change meaningfully? Explain briefly.

8. Conclusion

- a) Summarize your main empirical conclusion in 2–3 sentences.
- b) Give one caveat about interpreting panel FE estimates as causal in this setting.

Appendix

Data

File: temperature_growth_teaching.csv

Unit of Observation: Country-year

Coverage: 1960–2010 (annual)

Variable	Description
<i>iso3</i>	country code
<i>country</i>	country name
<i>region</i>	broad region
<i>year</i>	calendar year
<i>poor</i>	1 if baseline income below median
<i>gdp_growth</i>	annual log GDP per-capita growth
<i>log_gdppc</i>	log GDP per capita
<i>temp_c</i>	annual mean temperature in (°C)
<i>precip_mm</i>	annual precipitation in mm
<i>temp_dev</i>	temperature deviation from the country mean
<i>precip_dev</i>	precipitation deviation from the country mean
<i>temp_dev_poor</i>	$temp_dev \times poor$

References

Dell, M., Jones, B. F., & Olken, B. A. (2012). Temperature Shocks and Economic Growth: Evidence from the Last Half Century. *American Economic Journal: Macroeconomics*, 4(3), 66–95.